Engineering Curves: Construction of Epicycloid

Rolling Circle Method with Tangent & Normal

Gajavada Sanjeevkumar

September 8, 2026

Problem

A generating circle of diameter 30 mm rolls externally without slipping on a directing circle of diameter 90 mm. Construct one complete epicycloid arch corresponding to $\angle AOB = 120^\circ$, divide the arch into 12 equal angular parts, and retain every construction element in all subsequent views.

Given Data and Calculations

Given data

$$\text{Diameter of directing circle} = 90 \text{ mm},$$
$$R = 45 \text{ mm},$$
$$\text{Diameter of generating circle} = 30 \text{ mm},$$
$$r = 15 \text{ mm},$$
$$\angle AOB = 120^\circ,$$
$$\text{Let number of divisions} = 12.$$

Calculations

$$OC_0 = R + r = 45 + 15 = 60 \text{ mm},$$
$$AC_0 = r = 15 \text{ mm},$$
$$\Delta\theta = \frac{120^\circ}{12} = 10^\circ.$$

Generating-circle division

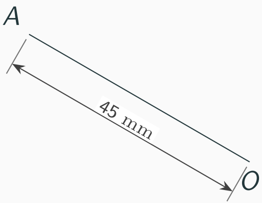
$$\text{Division angle} = \frac{360^\circ}{12} = 30^\circ.$$

Rolling relation

$$\text{Each } 10^\circ \text{ centre step} \leftrightarrow 30^\circ \text{ generating-circle step.}$$

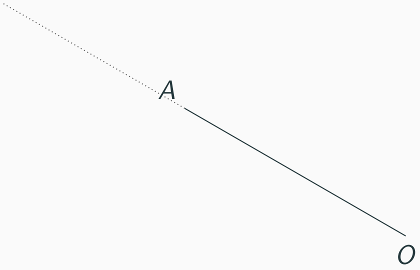
Step 1: Draw OA

$OA = 45$ mm.



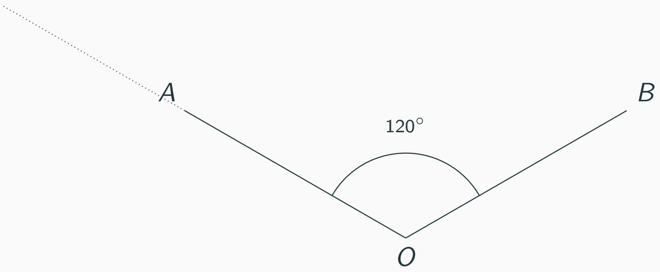
Step 2: Extend OA

Extend OA.



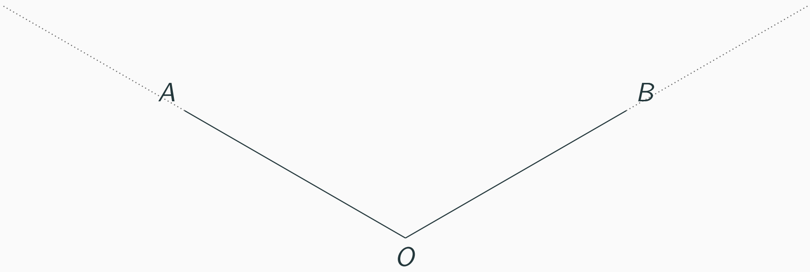
Step 3: Draw OB at 120°

$\angle AOB = 120^\circ$.



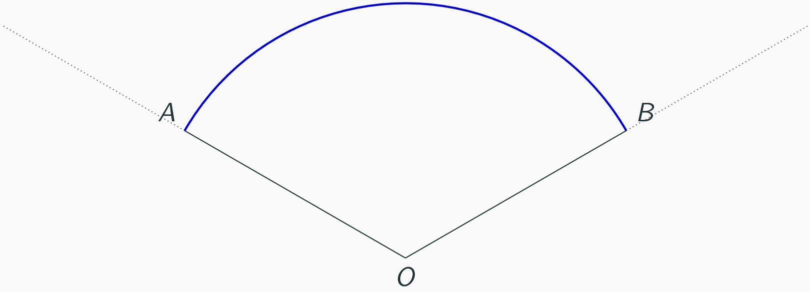
Step 4: Extend OB

Extend OB.



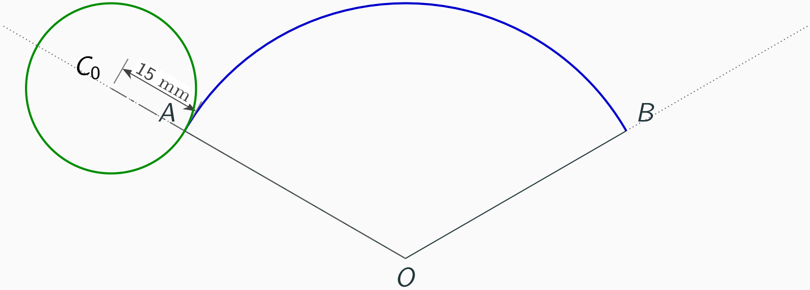
Step 5: Draw the directing arc AB

Blue arc AB: directing circle.



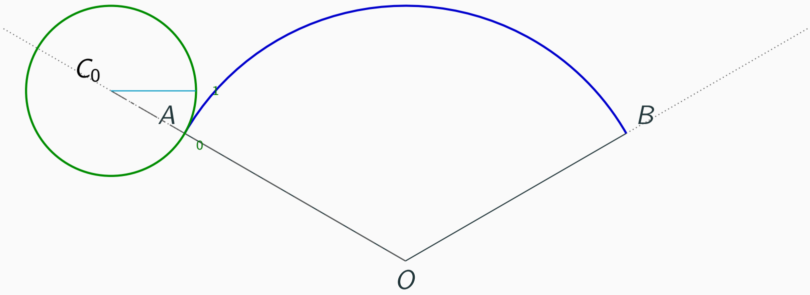
Step 6: Locate C_0 and the generating circle

$AC_0 = 15$ mm.
Green: generating circle.



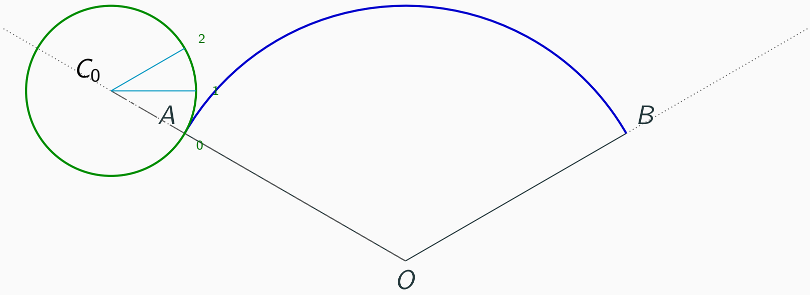
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



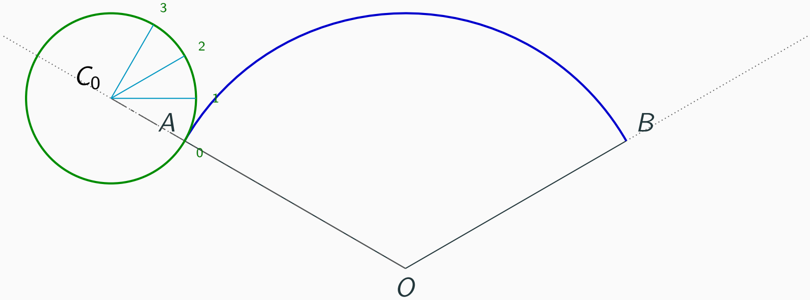
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



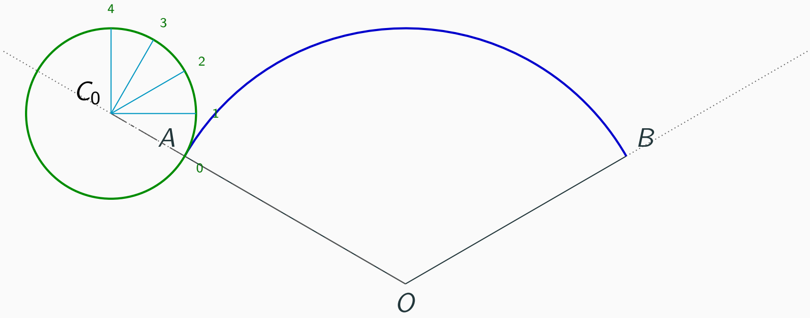
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



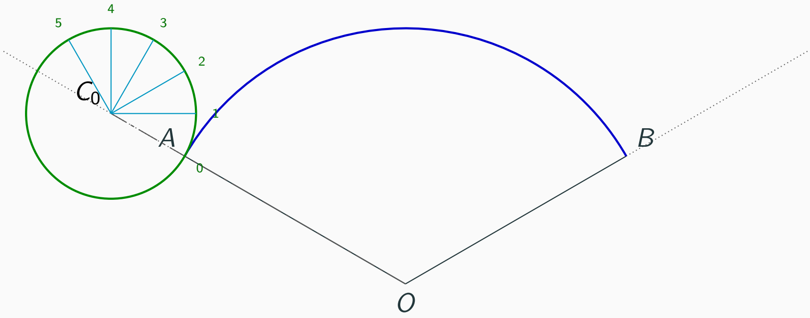
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



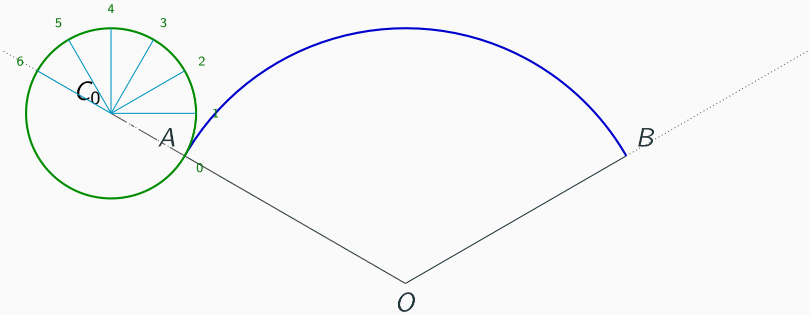
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



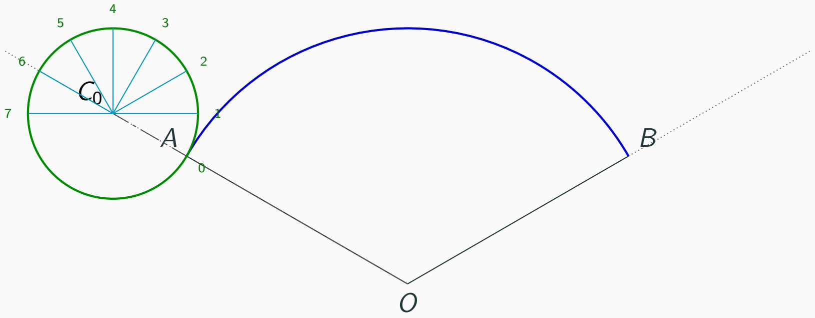
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



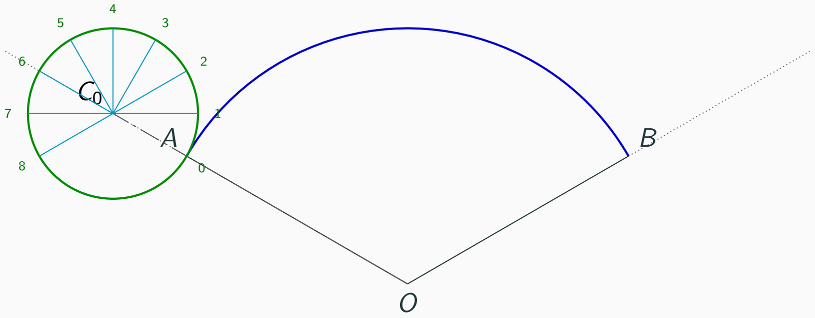
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



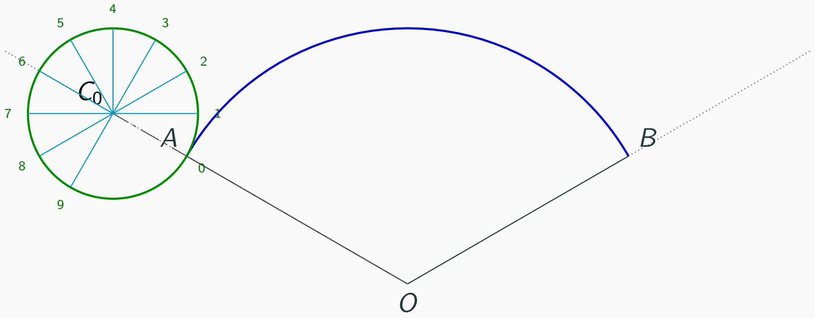
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



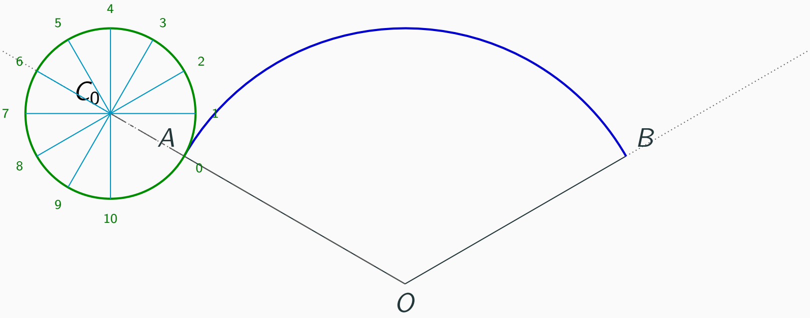
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



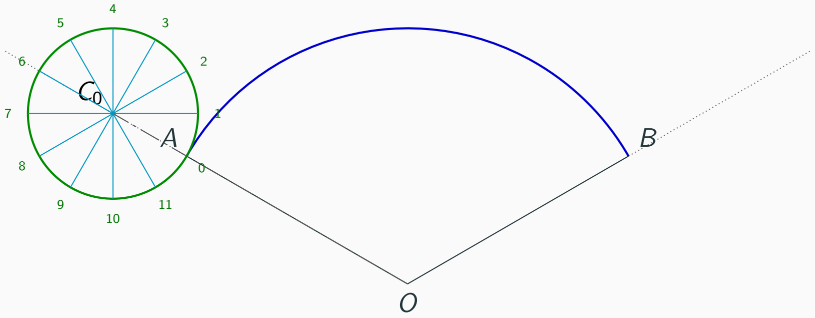
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



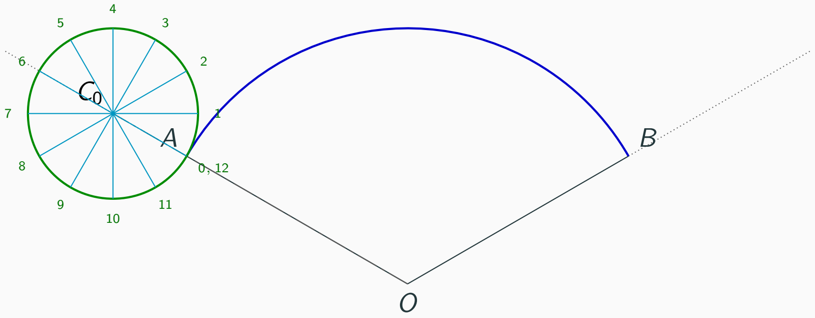
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



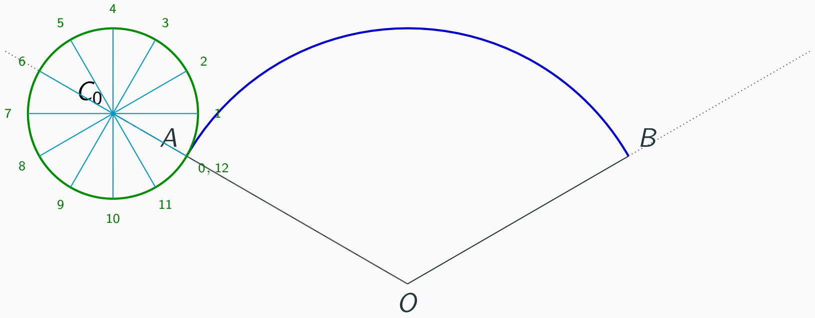
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



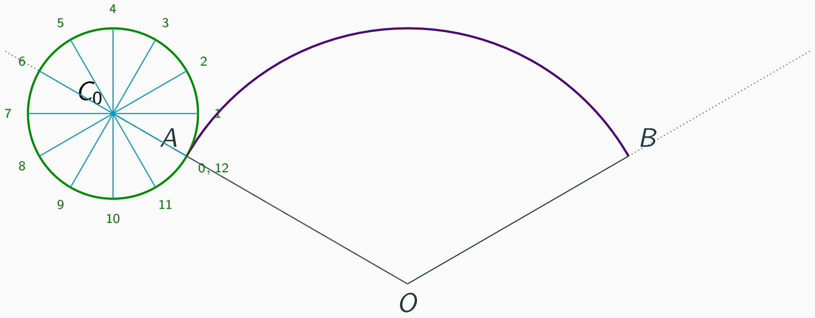
Step 8: Complete the generating-circle divisions

Generating-circle divisions.



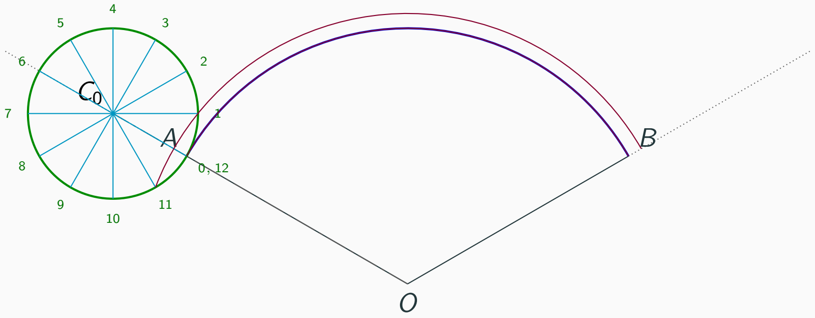
Step 9: Draw concentric construction arcs

Concentric construction arc.



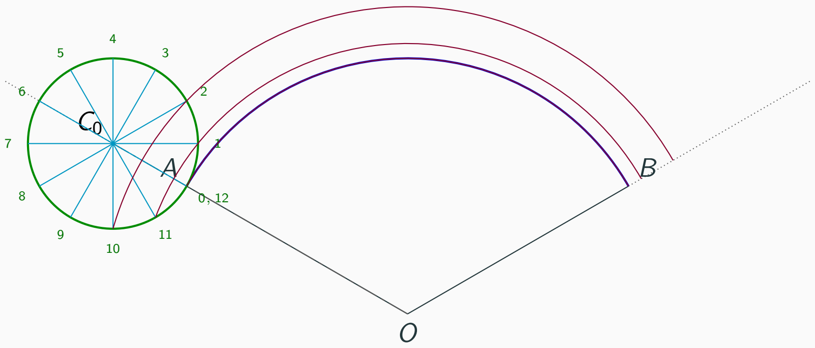
Step 9: Draw concentric construction arcs

Concentric construction arc.



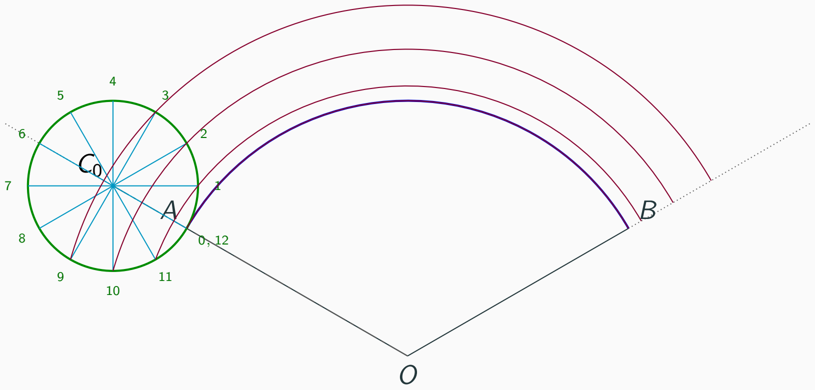
Step 9: Draw concentric construction arcs

Concentric construction arc.

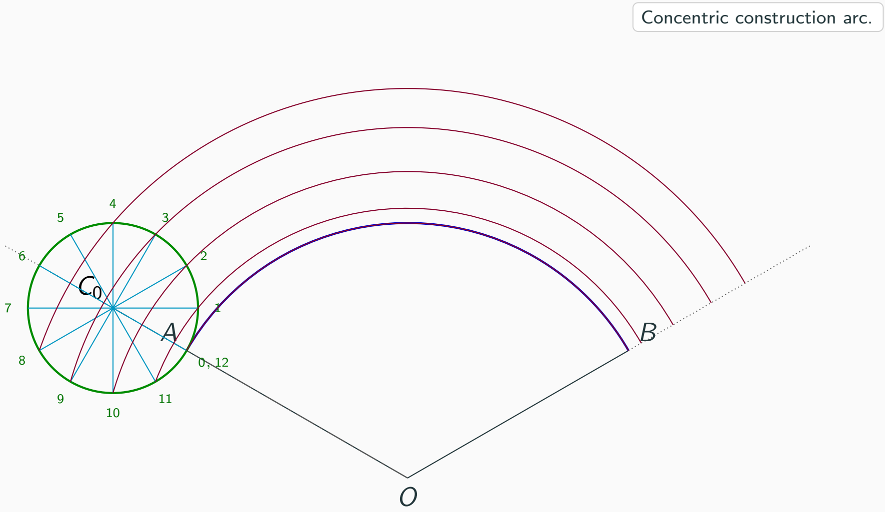


Step 9: Draw concentric construction arcs

Concentric construction arc.

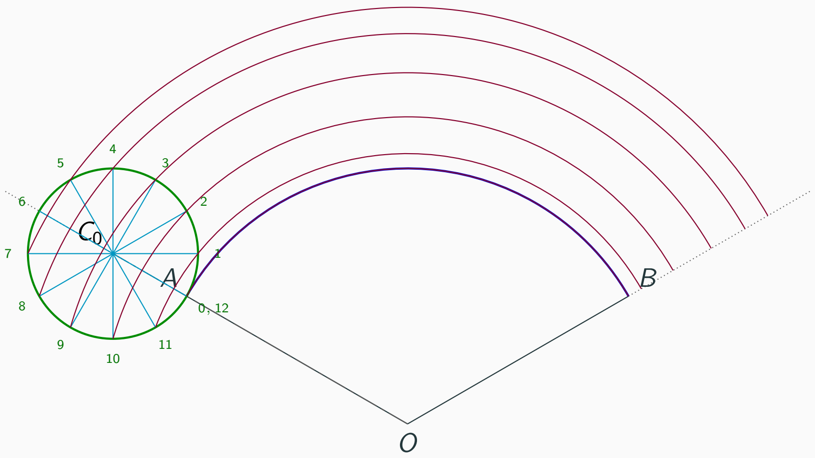


Step 9: Draw concentric construction arcs



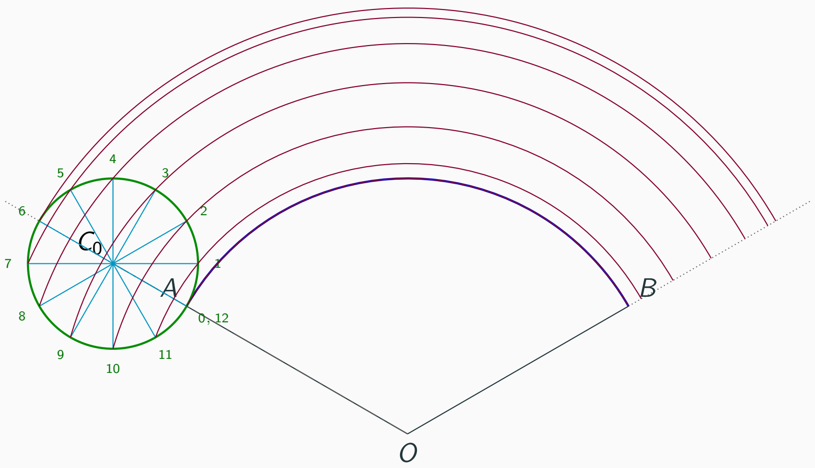
Step 9: Draw concentric construction arcs

Concentric construction arc.

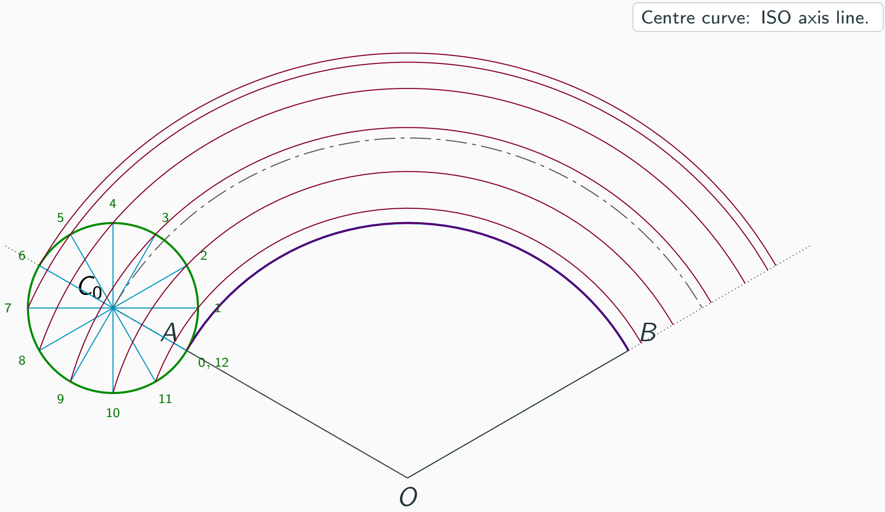


Step 9: Draw concentric construction arcs

Concentric construction arc.

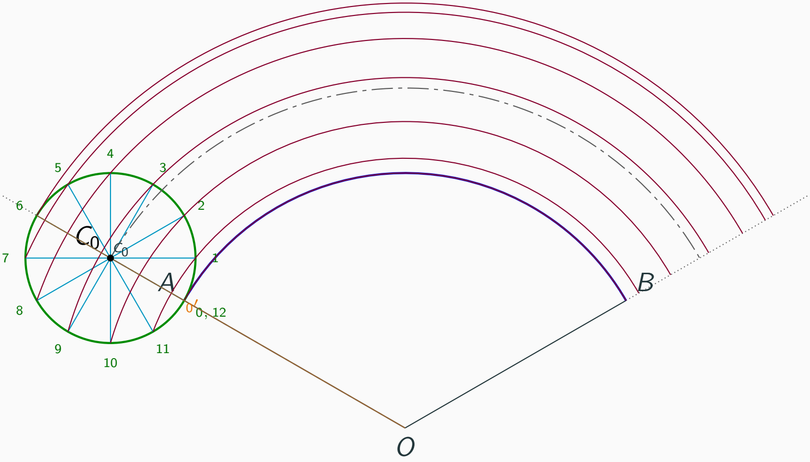


Step 10: Draw the centre arc through C_0



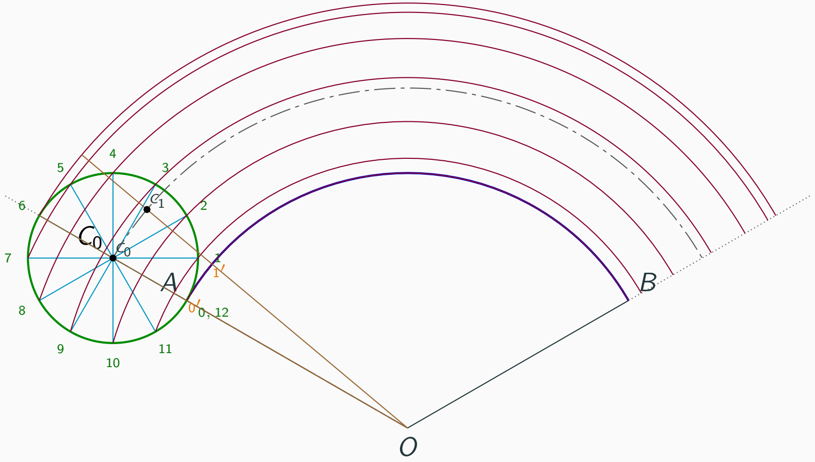
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



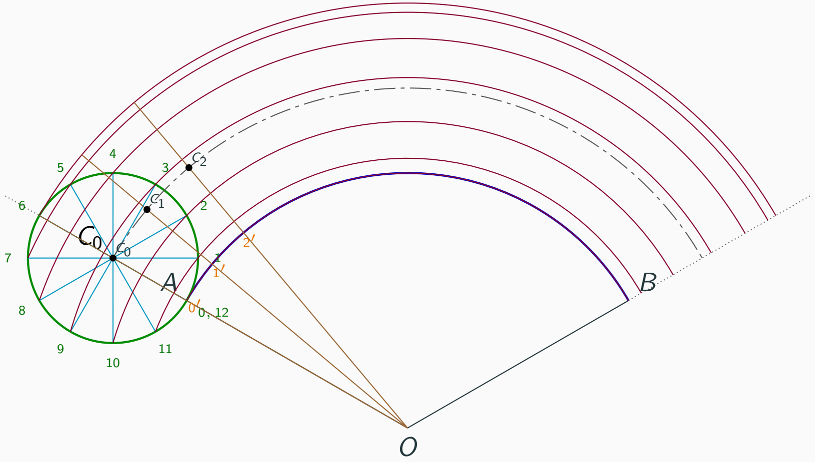
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.

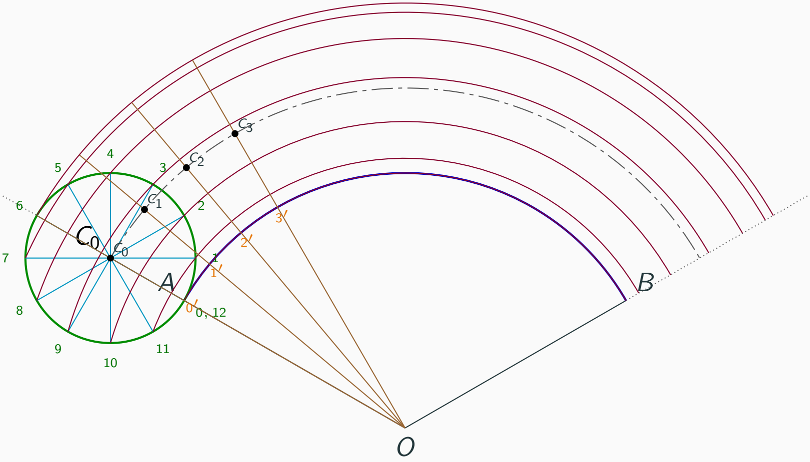


Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.

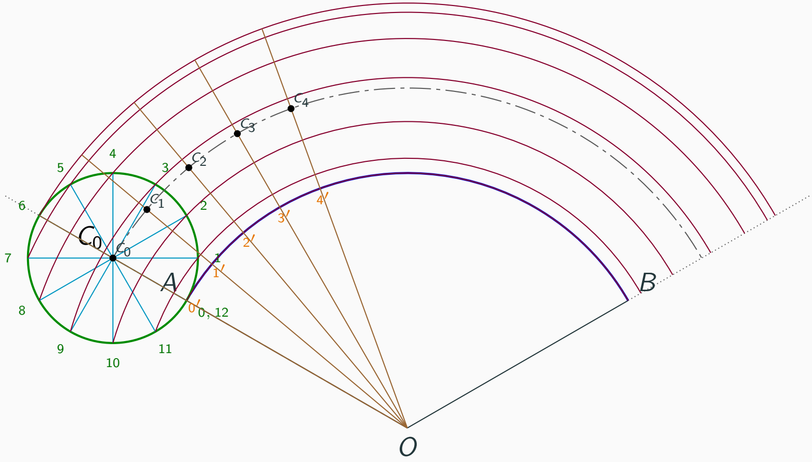


10° divisions.



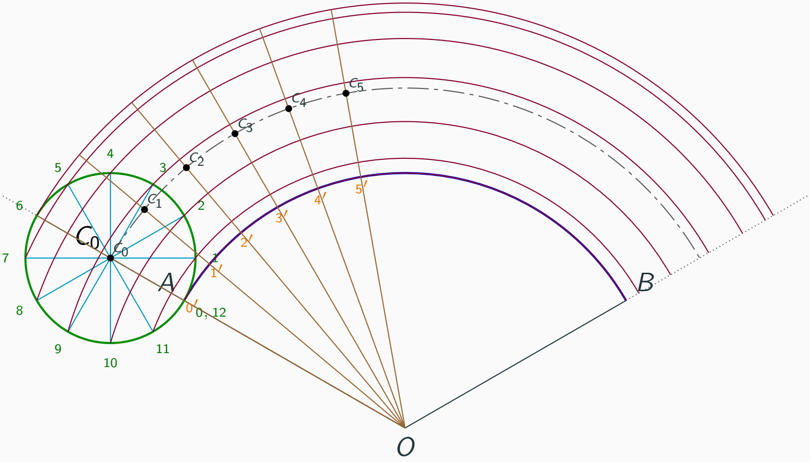
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



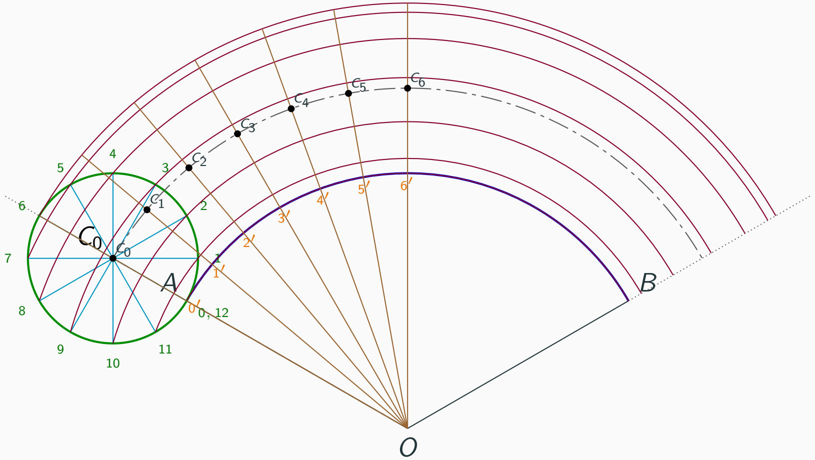
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



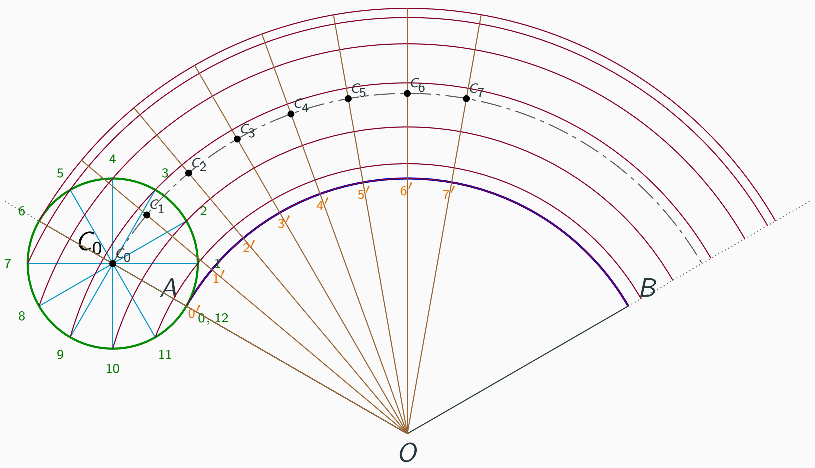
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.

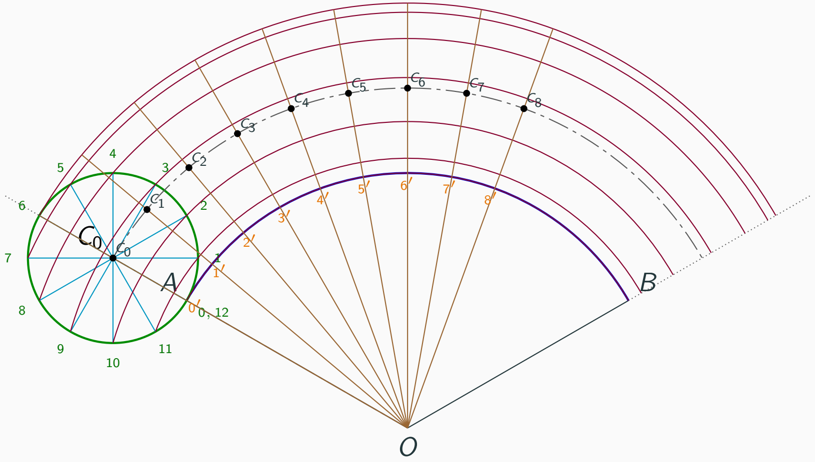


Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.

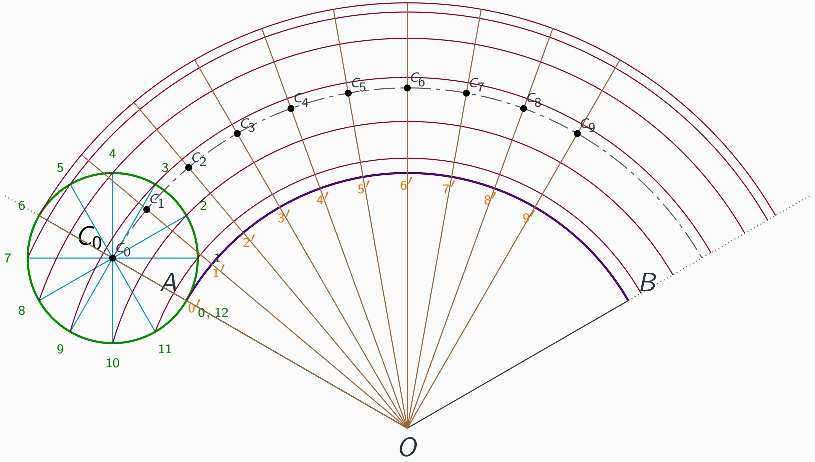


10° divisions.



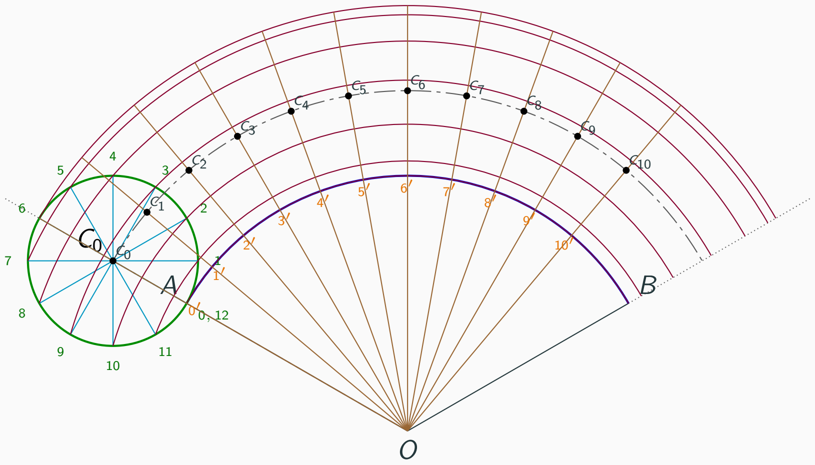
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



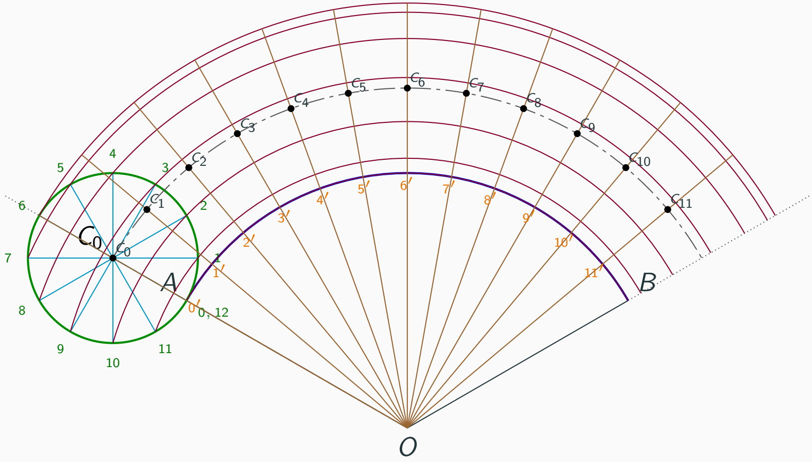
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



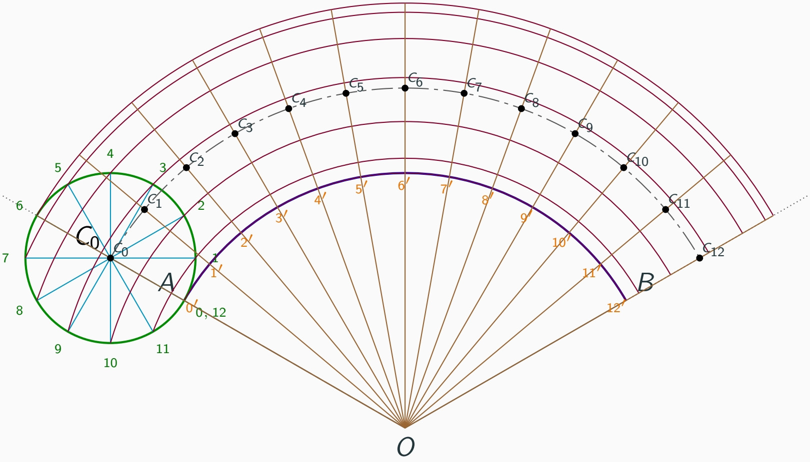
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



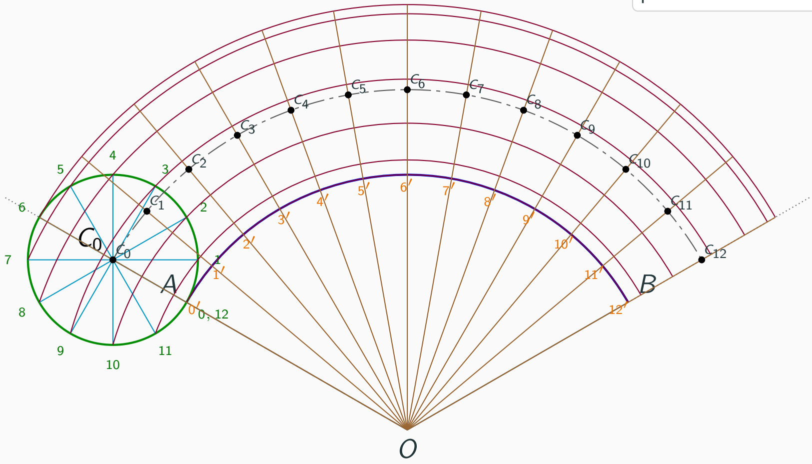
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



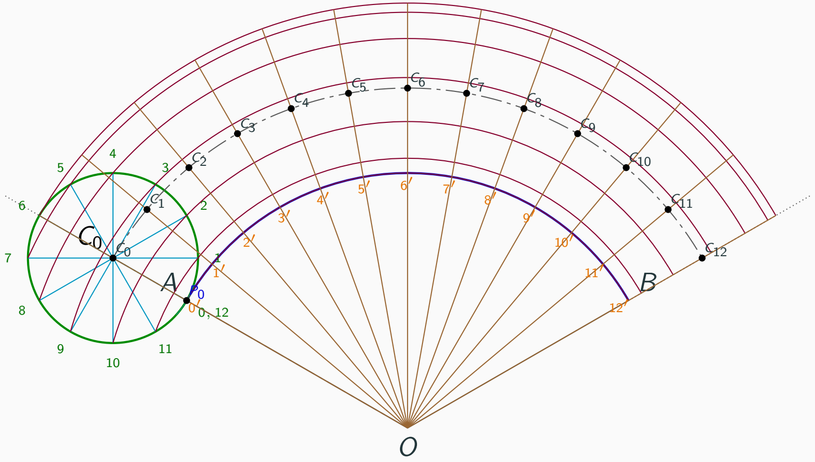
Step 12: Complete the angular divisions and points

Division points and centre points.

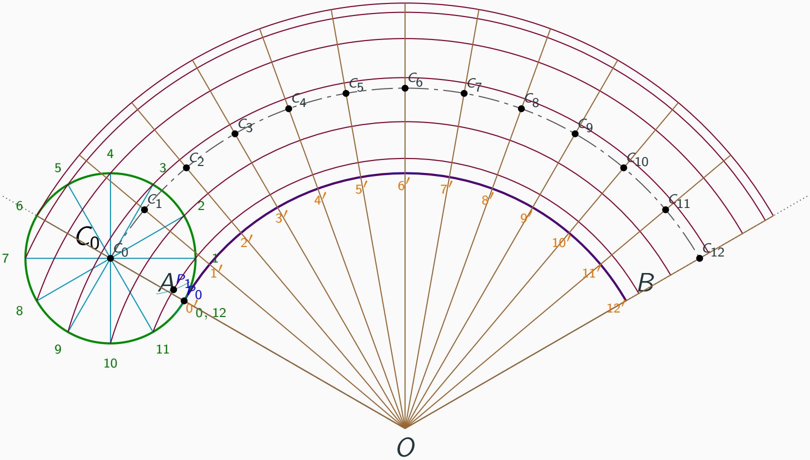


Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.

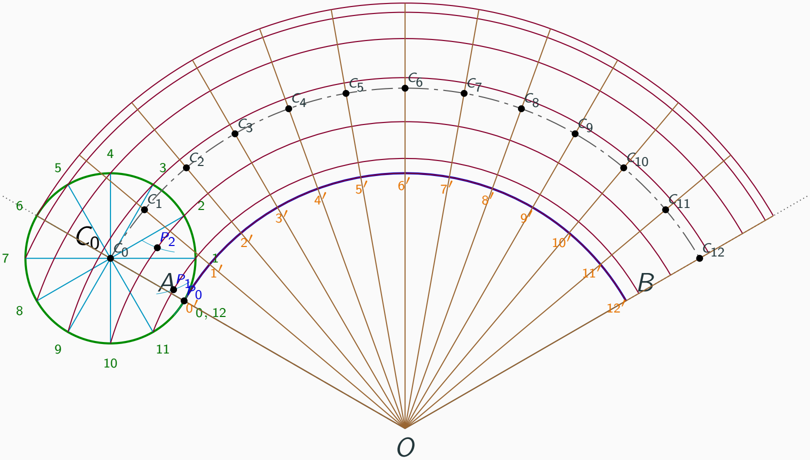


Cyan arc: radius 15 mm.



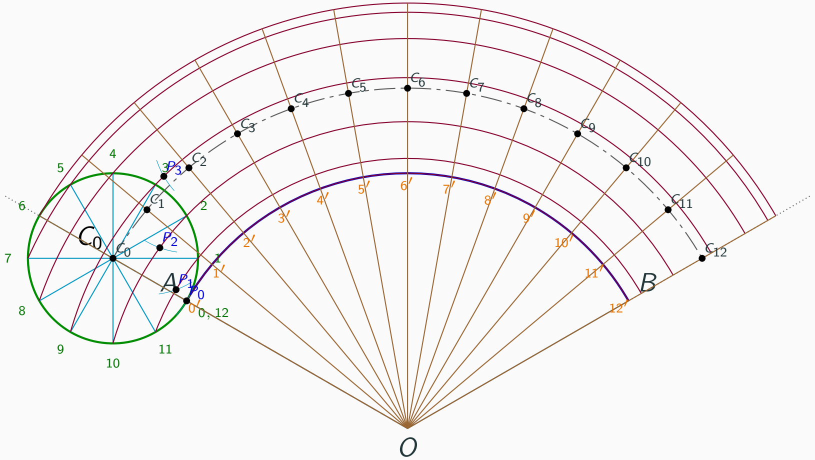
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



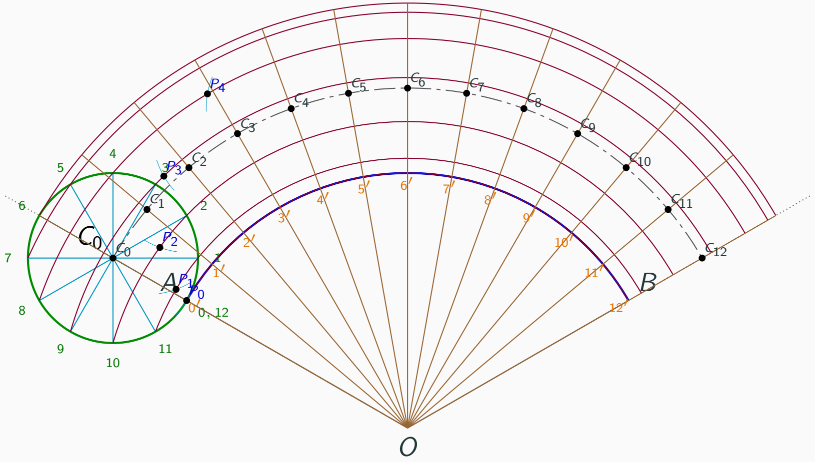
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



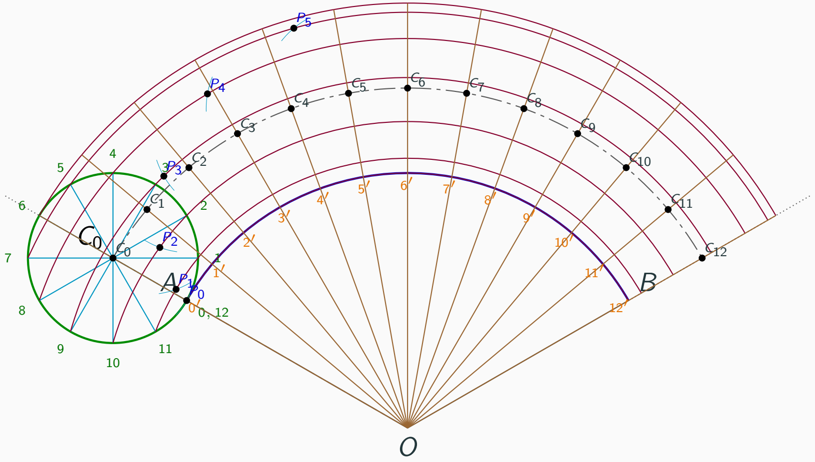
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



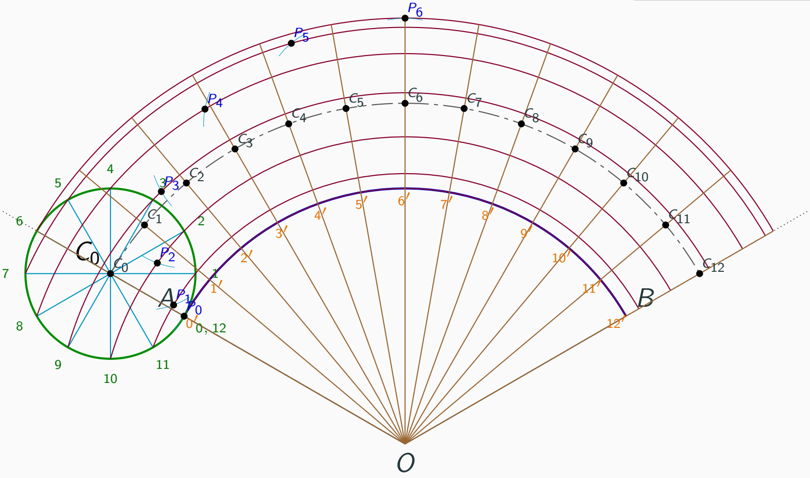
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



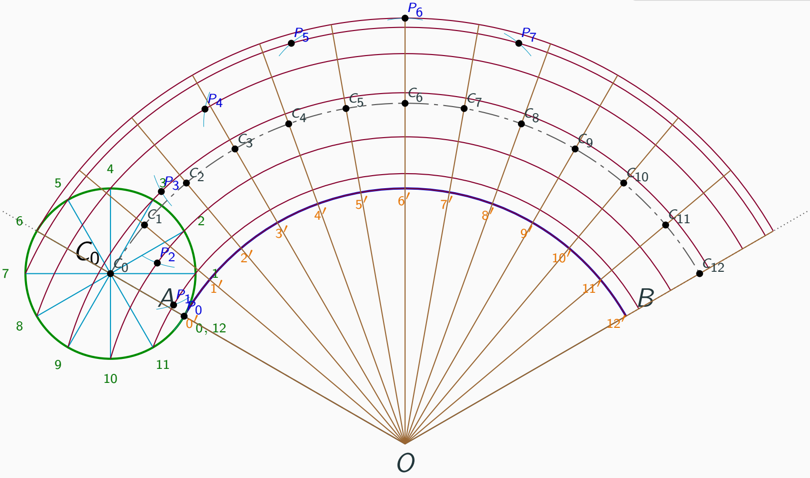
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



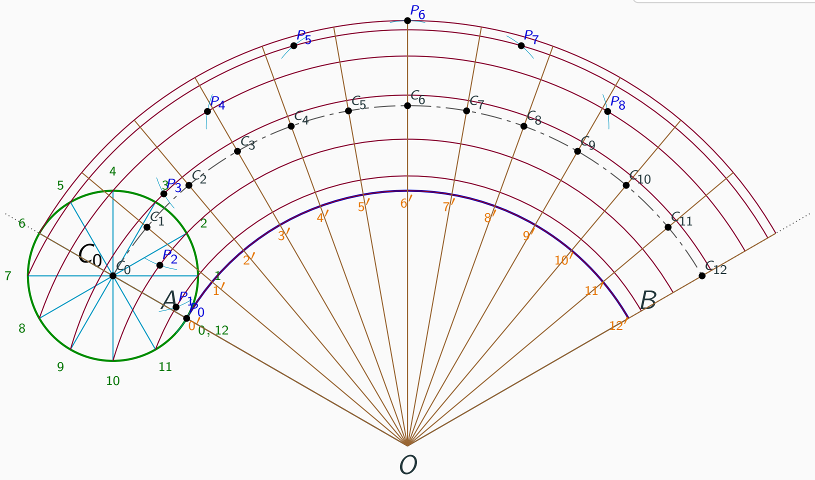
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



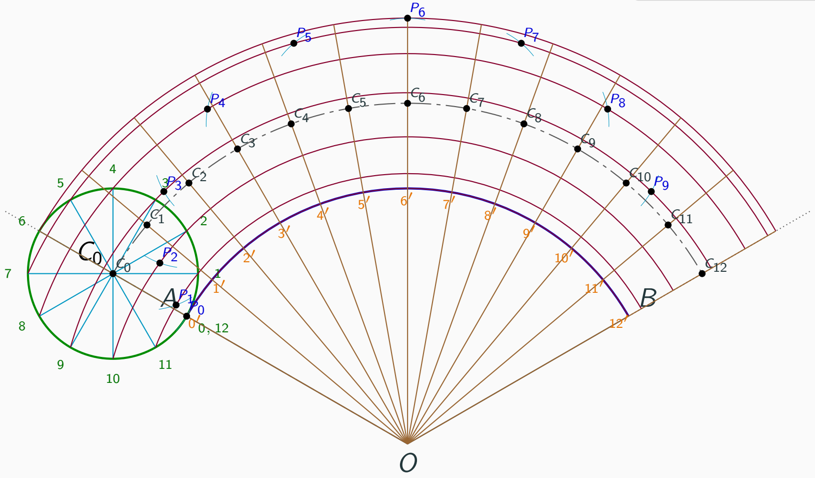
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



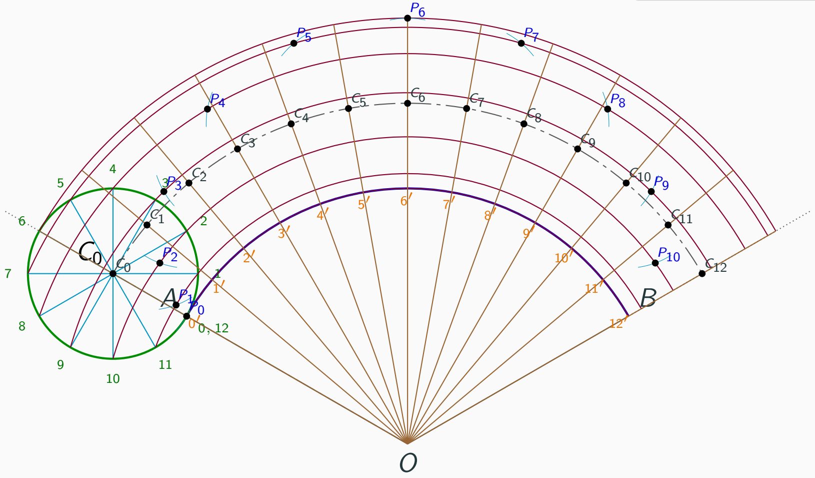
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



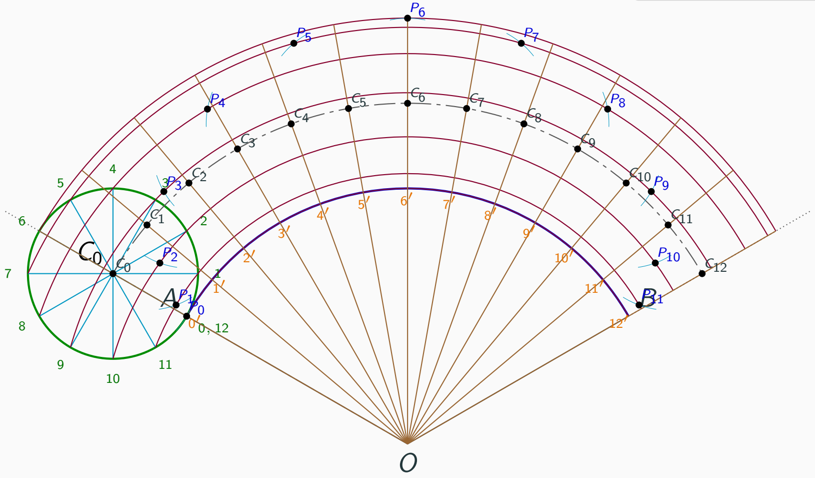
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



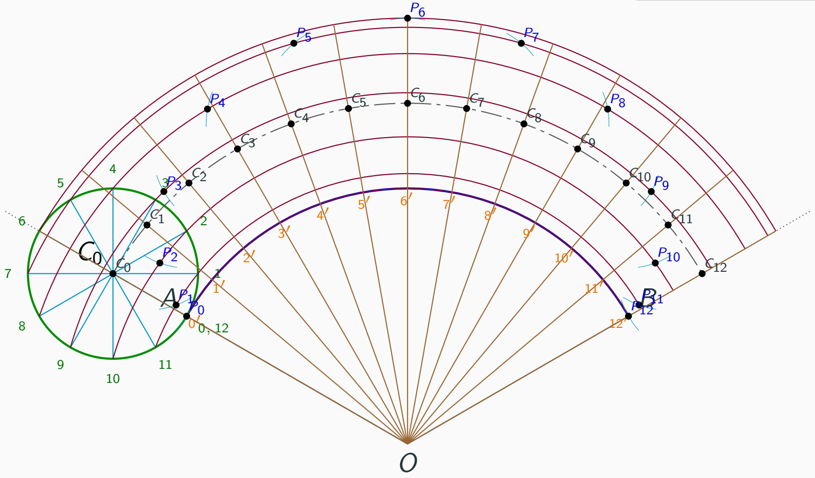
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



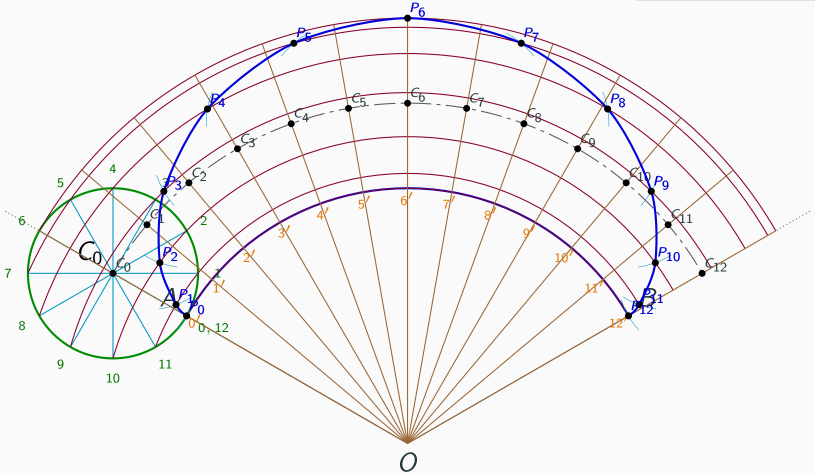
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



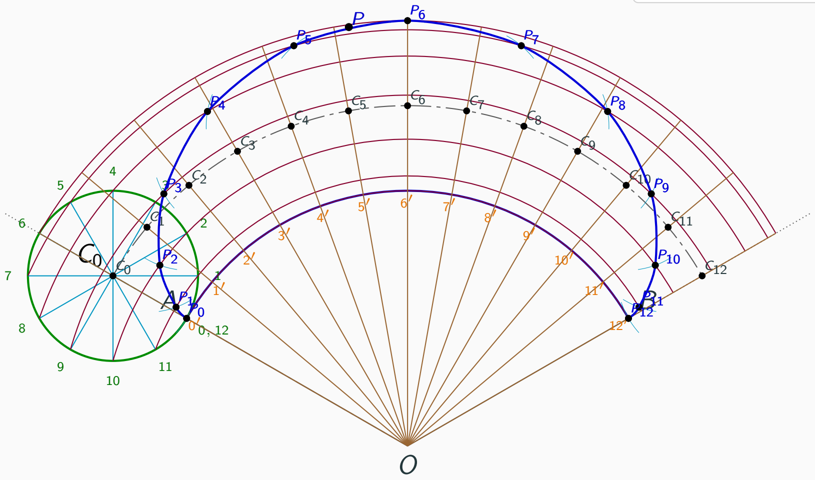
Step 14: Draw the epicycloid

Blue curve: epicycloid.



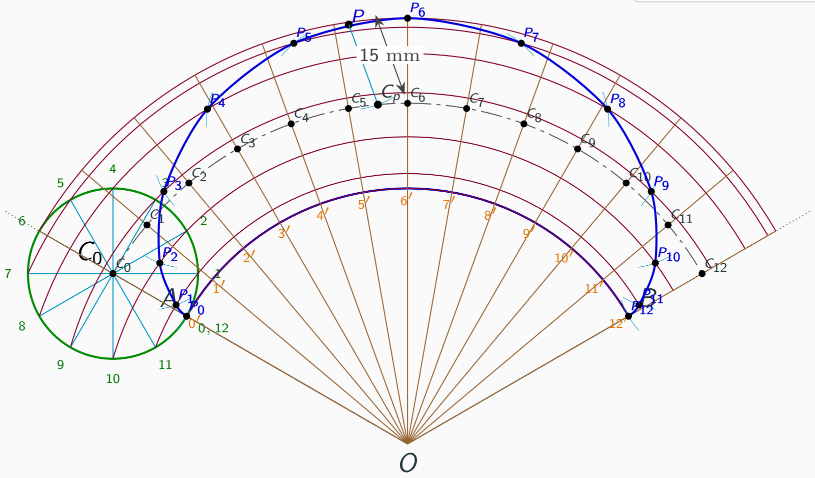
Step 15: Select a point P on the epicycloid

Selected point P.



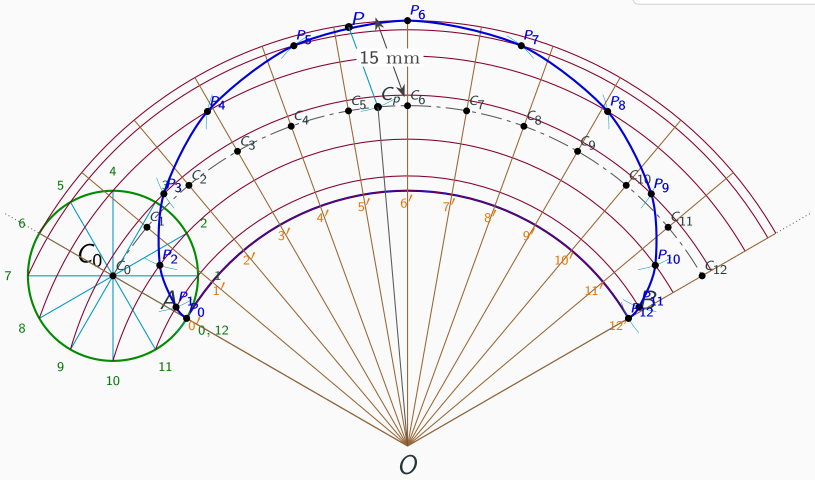
Step 16: Draw the 15 mm arc from P to the centre curve

$PC_p = 15\ mathrmmm.$



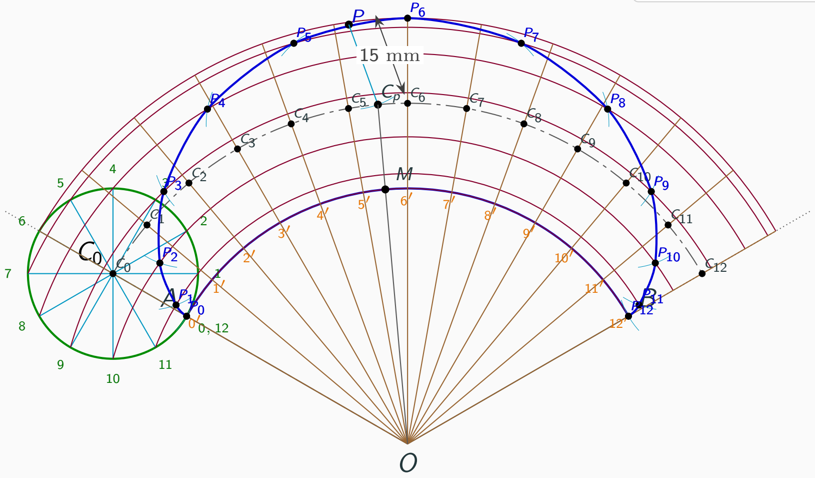
Step 17: Join C_p to O

Join C_p to O.

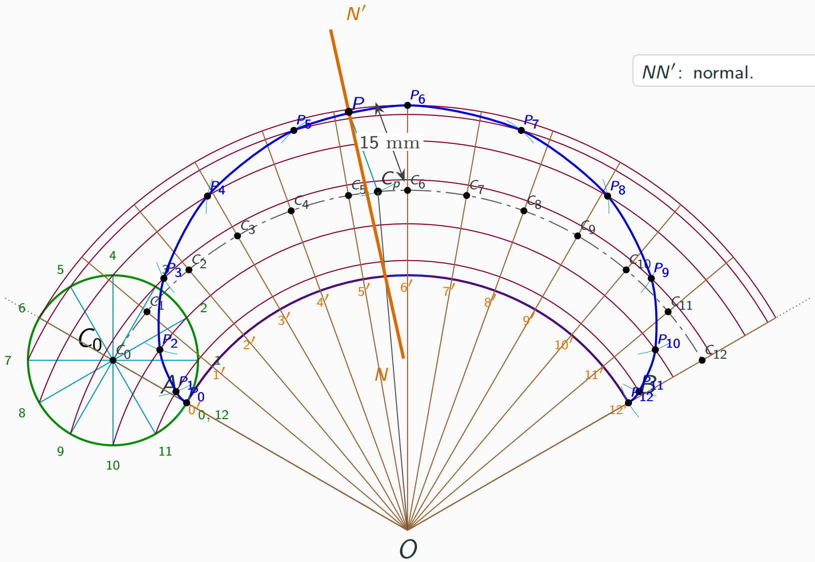


Step 18: Mark M on the directing circle

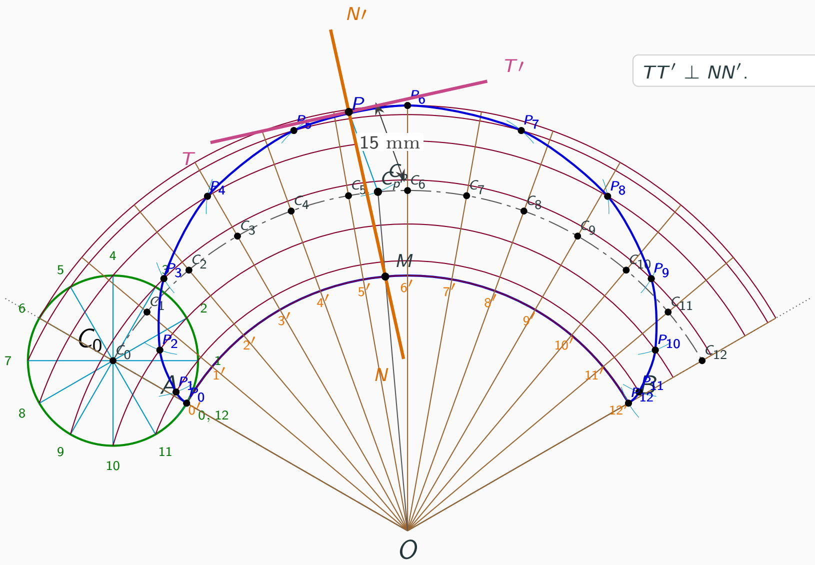
$M = OC_p \cap AB.$



Step 19: Draw the normal NN' through P

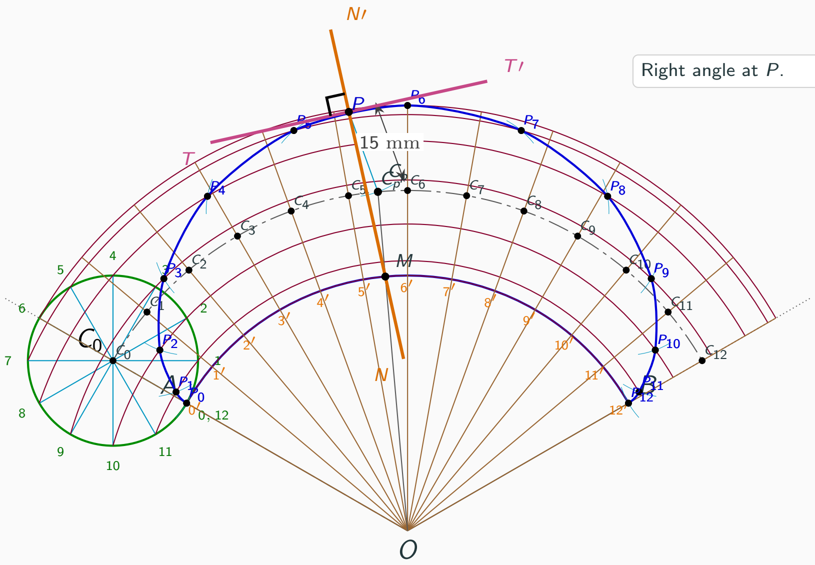


NN': normal.



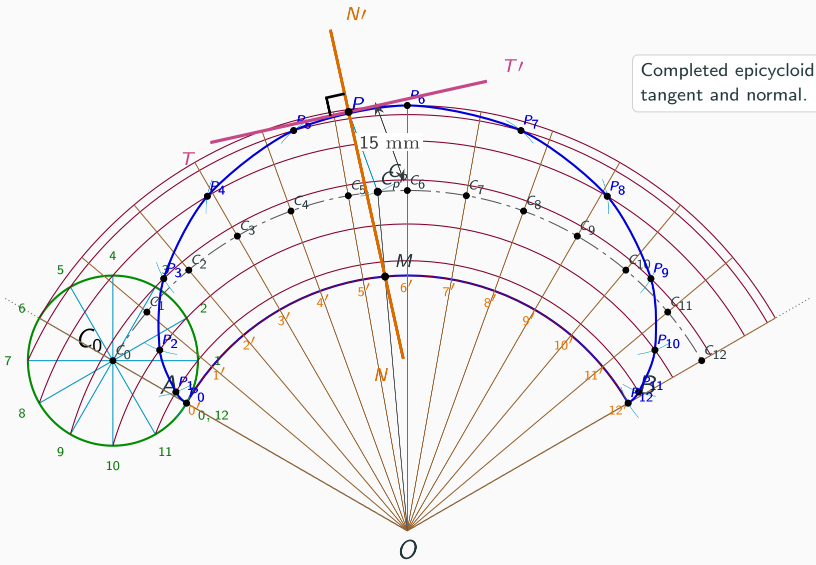
N'
T'
T
P
P6
P5
P7
P4
P8
P3
P9
P2
P10
P1
P0
P11
P12
15 mm
C5
C6
C7
C4
C8
C3
C9
C2
C10
C1
C11
C0
C12
M
N
A
B
O
1' 2' 3' 4' 5' 6' 7' 8' 9' 10' 11' 12'
0'
0, 12
1 2 3 4 5 6 7 8 9 10 11

$TT' \perp NN'$.



Right angle at P.

Step 20: Final completed epicycloid construction



Completed epicycloid with tangent and normal.